J. Vis. Commun. Image R. 35 (2016) 184–192

Contents lists available at ScienceDirect

J. Vis. Commun. Image R.

journal homepage: www.elsevier.com/locate/jvci

Image representation by harmonic transforms with parameters in $SL(2,R)$ ☆

Min Qi [a,c], Bing-Zhao Li [a,b,*], Huafei Sun [a,b]

[a] School of Mathematics and Statistics, Beijing Institute of Technology, Beijing 100081, China
[b] Beijing Key Laboratory on MCAACI, Beijing Institute of Technology, Beijing 100081, China
[c] Beijing College Finance and Commerce, Beijing 110000, China

ARTICLE INFO

Article history:
Received 4 August 2014
Accepted 15 December 2015
Available online 23 December 2015

Keywords:
Image representation
Orthogonal moments and transforms
2-D linear canonical transform
Polar harmonic linear canonical transform
2-D linear canonical transform series
Kernel
Transform coefficients
Parameter sensitivity

ABSTRACT

In this paper, a kind of invariant harmonic transforms with parameters in $SL(2,R)$ are proposed, which include the polar linear canonical transform (PLCT) and the two-dimensional linear canonical transform series (2-D LCTS). The capabilities of the PLCT and the 2-D LCTS on image representation are analyzed. The experimental results show that the 2-D LCTS has much stronger capability on the image representation with respect to characters, and has better invariance of scale and noise than the other transforms. Moreover, due to the varieties of parameters, the performance of the image representation is going bad when parameters used in the reconstructing process are inconsistent with those in the decomposing process. In other words, the proposed transforms can be used for the protection of the image safety because they have free parameters than the traditional methods.

1. Introduction

With the rapid development of the field of computer and communication technology, digital images have become a significant part of our lives. A large number of techniques for image processing and reconstruction have been proposed in recent years. Among them, the invariant moments and transforms have attracted more and more researchers, and successfully been applied in visual pattern recognition, face recognition, character recognition, digital watermarking and edge detection. Readers are encouraged to refer to [1–11] for more details and information.

Let $f(x,y)$ be a given image function, then the moments or transform coefficients are defined as

$$M_{nm} = \int\int_{x^2+y^2\leqslant 1} f(x,y)K^*_{nm}(x,y)dxdy,$$

where K_{nm} is a kernel with respect to integers n and m, and $*$ denotes the complex conjugate. Based on the orthogonality property of the kernel K_{nm}, the moments and transforms are divided into the orthogonal type and the non orthogonal type. The non-orthogonal type includes Fourier-Radial Mellin Descriptors (FRMD) [12], Rotational Moments (RMs) [13] and Complex Moments (CMs) [14,15]; while the orthogonal type includes Zernike Moments (ZMs) [16,17], Pseudo-Zernike Moments (PZMs) [13], Orthogonal Fourier–Mellin Moments (OFMMs) [3,18], Tchebichef Moments (TM) [19] and Radial-Harmonic Fourier Moments (RHFMs) [20]. Because of their good image-describing and image-reconstructing capability, these methods are widely applied in image processing.

In order to overcome the high computational expense and numerical instability issues at the greater moments of traditional moments and transforms, Yap, Jiang and Kot in [22] introduced a new kind of transforms, named the Polar Harmonic Transforms (PHTs), whose kernels are significantly simpler to compute compared with ZMs and PZMs. These transforms have advantages in time complexity and numerical stability, and have been applied in the image watermarking, seeing details in [23,24]. It should be noticed that the PHTs can be seen as the forms based on the theory of the Fourier Transform (FT) which is a special case of the Linear Canonical Transform.

The Linear Canonical Transform (LCT) was introduced in [25] by Moshinsky and Quesne during 1970s, which is a linear integral transform with parameters in a real special linear group $SL(2,R)$, and has been proved to be a powerful tool in several areas, including optics, signal processing, optimal filtering and time–frequency analysis. We encourage the interested reader to consult [26–28,30,29,31,32]. As will be seen that the FT, the fractional Fourier Transform (FrFT), and the Fresnel transform (FRT) are the

☆ This paper has been recommended for acceptance by Yehoshua Zeevi.

* Corresponding author at: School of Mathematics and Statistics, Beijing Institute of Technology, Beijing 100081, China.

E-mail address: li_bingzhao@bit.edu.cn (B.-Z. Li).

http://dx.doi.org/10.1016/j.jvcir.2015.12.010
1047-3203/

particular cases of the LCT. Until recently, to the best of our knowledge, there are few research papers published about the LCT on the image representation. Therefore, it is worthwhile to explore the image representation associated with the theory of LCT.

In this paper, a kind of harmonic transforms with parameters in a real special linear group $SL(2,R)$ are proposed to improve the capabilities of the existing results on the image representation and the image safety. The capabilities of the new transforms on the image representation are analyzed. Subsequently, the simulation are also presented to verify image representation of the new transforms, and the comparison of the proposed method with the existing results are also given.

2. Preliminary

2.1. Orthogonal moments and transforms

In this subsection, we will present the basic concepts of the orthogonal moments and transforms. For any integers n and m, the moments (or transform coefficients) M_{nm} of a finite function $f(x,y)$ is represented as [14,17,19,21]:

$$M_{nm} = \int_0^{2\pi}\int_0^1 f(r,\theta)K^*_{n'm'}(r,\theta)rdrd\theta,$$

with $K_{nm}(r,\theta) = R_n(r)A_m(\theta)$, where $R_n(r)$ and $A_m(\theta)$ denote the radial component and the angle component, respectively.

Compared with the non-orthogonal moments and transforms, the performance of image representation can be improved by the orthogonal moments and transforms. Thus, we pay more attention on the orthogonal moments and transforms which satisfy the orthogonality condition, i.e.,

$$\begin{aligned}\int_0^{2\pi}\int_0^1 K_{nm}(r,\theta)K^*_{nm}(r,\theta)rdrd\theta &= \int_0^1 R_n(r)R^*_{n'}(r)rdr \\ &\quad\times\int_0^{2\pi} A_m(\theta)A^*_{m'}(\theta)d\theta \\ &= c_1c_2\delta_{nn'}\delta_{mm'},\end{aligned}$$

where $\delta_{nn'} = 1$ if $n = n'$, otherwise $\delta_{nn'} = 0$ ($\delta_{mm'} = 1$ if $m = m'$, otherwise $\delta_{mm'} = 0$), and c_1, c_2 are constants.

Normally, the angle component is defined as $A_m(\theta) = \exp(im\theta)$, where $i^2 = -1$. Then different definitions of $R_n(r)$ are used to obtain different moments and transforms. For example, the Polar Complex Exponential Transform (PCET) in the PHTs is defined as [22]:

$$M_{nm} = \frac{1}{\pi}\int_0^{2\pi}\int_0^1 f(r,\theta)K^*_{nm}(r,\theta)rdrd\theta, \tag{1}$$

with $K_{nm}(r,\theta) = \exp(i2\pi nr^2)\exp(im\theta)$; and the Polar Cosine Transform (PCT) is given by

$$M^C_{nm} = \Omega_n\int_0^{2\pi}\int_0^1 f(r,\theta)K^{C*}_{nm}(r,\theta)rdrd\theta, \tag{2}$$

with $K^C_{nm} = \cos(\pi nr^2)\exp(im\theta)$, where $\Omega_n = \frac{1}{\pi}$ if $n = 0$, otherwise $\Omega_n = \frac{2}{\pi}$. In [22], Yap et al. showed that the PCT can be viewed as the best transform among the PHTs on the performance of the image representation. In addition, the kernel of the ZMs and the kernel of the PZMs can be respectively defined as

$$K^z_{nm} = \sum_{s=0}^{(n-|m|)/2}\frac{(-1)^s(n-s)!r^{n-2s}}{s!((n-|m|)/2-s)!(n-|m|-s)!}e^{im\theta}, \quad n = 0,1,\ldots,$$

and

$$K^{pz}_{nm} = \sum_{s=0}^{n-|m|}\frac{(-1)^s(2n+1-s)!r^{n-s}}{s!(n+|m|+1-s)!(n-|m|-s)!}e^{im\theta}, \quad n = 0,1,\ldots,$$

where $\frac{n-|m|}{2}$ is an integer and $|m| \leqslant n$.

2.2. LCT

For the sake of convenience, we denote a real special linear group as

$$SL(2,R) = \{A \in M(2,R)|\det(A) = 1\},$$

where $M(2,R)$ is the set of 2×2 real matrices. In the following, we always write the matrix

$$A = \begin{bmatrix} a & b \\ c & d\end{bmatrix}$$

as $A = [a,b;c,d]$ for convenience. For any $A \in SL(2,R)$, it is easy to see that its inverse $A^{-1} = [d,-b;-c,a]$ and the determinant $\det(A) = ad - bc = 1$.

The one-dimensional LCT (short for 1-D LCT) with parameter $A = [a,b;c,d] \in SL(2,R)$ of a function $f(x)$ is defined as [33]:

$$L^A_f(u) = \mathcal{L}_A[f](u) = \begin{cases}\int_R f(x)K(u,x)dx, & b \neq 0,\\ \sqrt{d}\exp\left(\frac{i}{2}cdu^2\right)f(du), & b = 0,\end{cases} \tag{3}$$

where $K(u,x) = \sqrt{\frac{1}{i2\pi b}}\exp\left\{\frac{i}{2b}[du^2 - 2ux + ax^2]\right\}$, $\mathcal{L}_A$ is the unitary LCT operator, and L^A_f is a LCT of f corresponding to parameter A.

We know consequently that the FT, the FrFT as well as the FRT are particular cases of the LCT. In fact, if we let a parameter $A = [\cos\theta, \sin\theta; -\sin\theta, \cos\theta]$, then the Eq. (3) can be represented as the following

$$\mathcal{L}_A[f](u) = \sqrt{\frac{1}{i2\pi\sin\theta}}\int_R\left[f(x)\exp\left\{\frac{i}{2}[\cot\theta u^2 - 2\csc\theta ux + \cot\theta x^2]\right\}\right]dx.$$

It implies that the LCT contains the FrFT. And if $A = [0,1;-1,0]$, the Eq. (3) simplifies to

$$\mathcal{L}_A[f](u) = \sqrt{\frac{-i}{2\pi}}\int_R[f(x)\exp\{-iux\}]dx,$$

which means that the LCT becomes to the product of the constant $\sqrt{-i}$ and the FT. Moreover, taking $A = [1,b;0,1]$, we can obtain the FRT as follows

$$\mathcal{L}_A[f](u) = \sqrt{\frac{1}{i2\pi b}}\int_R\left[f(x)\exp\left\{\frac{i}{2b}[u^2 - 2ux + x^2]\right\}\right]dx.$$

In the following, we will pay our attention to the LCT for $b \neq 0$.

Let $A = [a_1,b_1;c_1,d_1]$ and $B = [a_2,b_2;c_2,d_2]$, $b_1 \neq 0$, $b_2 \neq 0$. The authors in [38] gave the definition of two-dimensional LCT (short for 2-D LCT) with parameters A and B:

$$L^{A,B}_f(u,v) = \mathcal{L}_{A,B}[f](u,v) = \int_{R^2} f(x,y)K^{A,B}(u,v,x,y)dxdy, \tag{4}$$

where

$$\begin{aligned}K^{A,B}(u,v,x,y) = \frac{1}{2\pi}\sqrt{\frac{-1}{b_1b_2}}\exp\left\{\frac{i}{2b_1}[d_1u^2 - 2ux + a_1x^2]\right\}\\ \cdot\exp\left\{\frac{i}{2b_2}[d_2v^2 - 2vy + a_2y^2]\right\}.\end{aligned}$$

This implies that the 2-D LCT with matrix parameters in $SL(2,R)$ has eight parameters and its degree of freedom is 6. We also know that the two-dimensional FrFT can be reduced from the 2-D LCT when $A = B = [\cos\theta, \sin\theta; -\sin\theta, \cos\theta]$, and the two-dimensional FT can be obtained from the two-dimensional FrFT if $\theta = \frac{\pi}{2}$.

The inverse transform of the 2-D LCT (2-D ILCT) with parameters A^{-1} and B^{-1} is defined as:

$$\mathcal{L}_{A^{-1},B^{-1}}\{L^{A,B}_f\}(x,y) = \int_{R^2} L^{A,B}_f(u,v)K^{A^{-1},B^{-1}}(x,y,u,v)dudv, \tag{5}$$

where

$$K^{A^{-1},B^{-1}}(x,y,u,v)=\frac{1}{2\pi}\left(-\sqrt{\frac{-1}{b_1b_2}}\right)\exp\left\{\frac{-i}{2b_1}[a_1x^2-2xu+d_1u^2]\right\}\cdot\exp\left\{\frac{-i}{2b_2}[a_2y^2-2yv+d_2v^2]\right\},$$

and then $\mathcal{L}_{A^{-1},B^{-1}}\{\mathcal{L}_{A,B}[f]\}(x,y)=f(x,y)$.

At present, some well known aspects have been well researched and extended in the LCT domain, such as the uncertainty principles [34], the convolution and product theorem [35], the sampling theorem [36] and the fast algorithm [37]. However, there are few published paper about the image representation associated with the theory of LCT. Inspired by the moments and transforms and based on the theory of LCT, we will propose harmonic transforms with parameters in $SL(2,R)$: the polar linear canonical transform (short for PLCT) and the 2-dimensional linear canonical transform series (short for 2-D LCTS) in next section.

3. Harmonic Transforms with parameters in $SL(2,R)$

3.1. PLCT

Let $A=[a,b;c,d]$ be an element in $SL(2,R)$. The polar linear canonical transform (PLCT) is defined as:

$$M_{nm}^{\text{PLCT}}=\frac{1}{\pi}\int\int_D f(r,\theta)K_{nm}^{A*}(r,\theta)rdrd\theta, \tag{6}$$

with

$$K_{nm}^{A}(r,\theta)=R_n^A(r)\exp(im\theta)=\exp\left\{\frac{-i}{2b}[d(2\pi nb)^2-2r^2(2\pi nb)+ar^4]\right\}\exp(im\theta), \tag{7}$$

where $D=\{(r,\theta)|r\in[0,1],\theta\in[0,2\pi]\}$.

We have the following result.

Theorem 3.1. *The kernel $K_{nm}^A(r,\theta),(r,\theta)\in[0,1]\times[0,2\pi]$, of the PLCT given by Eq. (7) is orthogonal and complete.*

Proof. By simple computation, we have

$$\int_0^1 R_n^A(r)R_{n'}^{A*}(r)rdr=\frac{1}{2}\delta_{nn'},$$

and

$$\int_0^{2\pi}\int_0^1 K_{nm}^A(r,\theta)K_{n'm'}^{A*}(r,\theta)rdrd\theta=\pi\delta_{nn'}\delta_{mm'}.$$

It means that the kernel $K_{nm}^A(r,\theta)$ satisfies orthogonality condition. To prove the completeness, for any function f, suppose $\int_0^{2\pi}\int_0^1 f(r,\theta)K_{nm}^{A*}(r,\theta)rdrd\theta=0$. Since it is well known that the series $\{\exp(ik\theta),\ |k|=0,1,\ldots,\infty\}$ is complete in $\theta\in[0,2\pi]$, we have $\int_0^1 f\exp\left\{\frac{i}{2b}[4r^2\pi nb-d(2\pi nb)^2-ar^4]\right\}rdr=0$. What's more, the series $\{\exp(i2\pi nr^2),\ |n|=0,1,\ldots,\infty\}$ is also complete at $r\in[0,1]$. We immediately obtain that $f\equiv 0$. Therefore, the kernel $K_{nm}^A(r,\theta)$ of the PLCT is complete and orthogonal at $[0,1]\times[0,2\pi]$. □

Meanwhile, since the kernel of the PLCT is a basic wave, we name it harmonic transform. It is worthy noting that when choosing parameter $A=[0,b;-1/b,0]$, the PLCT becomes to the special case PCET. This suggests that the PLCT is a generalization of the PCET. In addition, the kernel of the PLCT has the form $\exp\left\{\frac{i}{2b}\left[d(2\pi nb)^2+ar^4\right]\right\}$, which can be viewed as the offset form and may affect the radial component and the angle component, especially the factor $\frac{a}{b}$. In the simulation section, we need to examine the capabilities of the PLCT on image reconstruction and analyze the effect of the parameters of the PLCT via simulations in this paper.

3.2. 2-D LCTS

We begin with the following auxiliary lemma.

Let $\mathcal{D}=\{(x,y)||x|>T_x/2,|y|>T_y/2,T_x,T_y$ are both positive *real numbers*$\}$. We call $f(x,y)$ is a finite function if $f(x,y)=0$ for any $(x,y)\in\mathcal{D}$.

Lemma 3.2. *Let $L_f^{A,B}(u,v)$ be a two-dimensional linear canonical transform function of a finite function $f(x,y)$, then $f(x,y)$ can be reconstructed from samples of its $L_f^{A,B}(u,v)$.*

Proof. By the modulation frequency shifting relation, the 2-D LCT with parameters $A=[a_1,b_1;c_1,d_1]$ and $B=[a_2,b_2;c_2,d_2]$ is given by

$$\mathcal{L}_{A,B}[f(x,y)\exp(iw_1x+iw_2y)]=\exp\left\{i\left(w_1d_1u-\frac{w_1^2b_1d_1}{2}\right)\right\}\cdot\exp\left\{i\left(w_2d_2v-\frac{w_2^2b_2d_2}{2}\right)\right\}\cdot L_f^{A,B}(u-w_1b_1,v-w_2b_2),$$

where w_1 and w_2 are real numbers.

Let $L_{T_u,T_v}^{A,B}(u,v)$ be the samples of $L_f^{A,B}(u,v)$. Then, it can be written as

$$L_{T_u,T_v}^{A,B}(u,v)=L_f^{A,B}(u,v)\sum_{m,n}\delta(u-nT_u)\delta(v-mT_v)=L_f^{A,B}(u,v)\frac{w_uw_v}{4\pi^2}\sum_{m,n}\exp[inw_u+imw_v], \tag{8}$$

where T_u, T_v are the sampling intervals, $w_u=\frac{2\pi}{T_u}$, $w_v=\frac{2\pi}{T_v}$, $\delta(u,v)$ is the two-dimensional Delta function, and $|n|,|m|=0,1,\ldots,\infty$. From Eq. (5), the sampled function $L_{T_u,T_v}^{A,B}(u,v)$ of the 2-D ILCT can be rewritten as

$$\hat{f}(x,y)=\mathcal{L}_{A^{-1},B^{-1}}\{L_{T_u,T_v}^{A,B}(u,v)\}=\frac{w_uw_v}{4\pi^2}\sum_{m,n}\exp\left\{i\left(nw_ua_1x+\frac{(nw_u)^2a_1b_1}{2}\right)\right\}\cdot\exp\left\{i\left(mw_va_2y+\frac{(mw_v)^2a_2b_2}{2}\right)\right\}f(x+nw_ub_1,y+mw_vb_2).$$

Thus, if the original function $f(x,y)$ vanishes for $|x|>\frac{|b_1|w_u}{2}$, $|y|>\frac{|b_2|w_v}{2}$, then f can be reconstructed from the term $n=m=0$, i.e., $\hat{f}(x,y)=\frac{w_uw_v}{4\pi^2}f(x,y)$ for $|x|\leqslant\frac{|b_1|w_u}{2},|y|\leqslant\frac{|b_2|w_v}{2}$.

Therefore, any finite function $f(x,y)(\{(x,y)||x|\leqslant T_x/2,|y|\leqslant T_y/2,T_x>0,T_y>0\})$ can be reconstructed from the sampled function of its 2-D LCT $L_f^{A,B}(u,v)$ at points $(u_n,v_m)=(nT_u,mT_v)$, if $T_u\leqslant\frac{2\pi|b_1|}{T_x},T_v\leqslant\frac{2\pi|b_2|}{T_y}$. Thus, the reconstruction formula of $f(x,y)$ can be given as

$$f(x,y)=\frac{4\pi^2}{w_uw_v}rect\left(\frac{x}{T_x}\right)rect\left(\frac{y}{T_y}\right)\mathcal{L}_{A^{-1},B^{-1}}\{L_{T_u,T_v}^{A,B}(u,v)\}(x,y).\quad\square \tag{9}$$

Now, we present the main result of this section.

Theorem 3.3. *For any finite function $f(x,y)$, its 2-D LCTS expansion with parameters A and B both $\in SL(2,R)$ can be derived as*

$$f(x,y) = \sum_{n,m} C^{A,B}_{nm} \sqrt{\frac{-1}{T_1T_2}} \times \exp\left\{\frac{-i}{2b_1}\left[d_1\left(n\frac{2\pi b_1}{T_1}\right)^2 - 2x\left(n\frac{2\pi b_1}{T_1}\right) + a_1x^2\right]\right\} \cdot \exp\left\{\frac{-i}{2b_2}\left[d_2\left(m\frac{2\pi b_2}{T_2}\right)^2 - 2y\left(m\frac{2\pi b_2}{T_2}\right) + a_2y^2\right]\right\}, \quad (10)$$

with the 2-D LCT expansion coefficients $C^{A,B}_{nm}$ defined as

$$C^{A,B}_{nm} = \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}}\int_{-\frac{T_2}{2}}^{\frac{T_2}{2}} f(x,y)\left(-\sqrt{\frac{-1}{T_1T_2}}\right) \cdot \exp\left\{\frac{i}{2b_1}\left[d_1\left(n\frac{2\pi b_1}{T_1}\right)^2 - 2x\left(n\frac{2\pi b_1}{T_1}\right) + a_1x^2\right]\right\} \cdot \exp\left\{\frac{i}{2b_2}\left[d_2\left(m\frac{2\pi b_2}{T_2}\right)^2 - 2y\left(m\frac{2\pi b_2}{T_2}\right) + a_2y^2\right]\right\}dxdy, \quad (11)$$

where $(x,y) \in [-\frac{T_1}{2}, \frac{T_1}{2}] \times [-\frac{T_2}{2}, \frac{T_2}{2}]$, $A = [a_1, b_1; c_1, d_1]$ and $B = [a_2, b_2; c_2, d_2]$.

Proof. In order to construct the completely unit orthogonal series related to the 2-D LCT, we first compute the 2-D ILCT of the impulse $\delta(u - nu_0, v - mv_0)$ in the 2-D LCT domain, where $u_0,\ v_0 \in R,\ |n|, |m| = 0, 1, \ldots, \infty$. We can find that

$$L_{A^{-1},B^{-1}}(\delta(u - nu_0, v - mv_0)) = \frac{-i}{2\pi\sqrt{b_1b_2}} \times \exp\left\{-\frac{i}{2b_1}[a_1x^2 - 2x(nu_0) + d_1(nu_0)^2]\right\} \cdot \exp\left\{\frac{-i}{2b_2}[a_2y^2 - 2y(mv_0) + d_1(mv_0)^2]\right\}.$$

Let

$$\tilde{\Phi}^{A,B}_{nm}(x,y) = \frac{-i}{2\pi\sqrt{b_1b_2}} \exp\left\{\frac{-i}{2b_1}[a_1x^2 - 2x(nu_0) + d_1(nu_0)^2]\right\} \cdot \exp\left\{\frac{-i}{2b_2}[a_2y^2 - 2y(mv_0) + d_1(mv_0)^2]\right\} \quad (12)$$

be the orthogonal system of the 2-D LCTS. We will calculate the inner product between any two functions of the orthogonal system along the interval $[-\frac{T_1}{2}, \frac{T_1}{2}] \times [-\frac{T_2}{2}, \frac{T_2}{2}]$. If $n = n', m = m'$, we have

$$\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}}\int_{-\frac{T_2}{2}}^{\frac{T_2}{2}} \tilde{\Phi}^{A,B}_{nm}(x,y)\tilde{\Phi}^{A,B*}_{n'm'}(x,y)dxdy = \frac{T_1T_2}{4\pi^2b_1b_2}.$$

If $n \neq n',\ m \neq m'$, we take $u_0 = \frac{2\pi k_1b_1}{T_1}$ and $v_0 = \frac{2\pi k_2b_2}{T_2}$, where $k_1,\ k_2$ are integers. To ensure the orthogonality of the kernel, we have

$$\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}}\int_{-\frac{T_2}{2}}^{\frac{T_2}{2}} \tilde{\Phi}^{A,B}_{nm}(x,y)\tilde{\Phi}^{A,B*}_{n'm'}(x,y)dxdy = 0.$$

By simple calculations, we obtain the unit orthogonal function of the 2-D LCT:

$$\Phi^{A,B}_{nm}(x,y) = \frac{-i}{\sqrt{T_1T_2}} \times \exp\left\{-\frac{i}{2b_1}\left[a_1x^2 - 2x\left(n\frac{2\pi k_1b_1}{T_1}\right) + d_1\left(n\frac{2\pi k_1b_1}{T_1}\right)^2\right]\right\} \cdot \exp\left\{-\frac{i}{2b_2}\left[a_2y^2 - 2y\left(m\frac{2\pi k_2b_2}{T_2}\right) + d_1\left(m\frac{2\pi k_2b_2}{T_2}\right)^2\right]\right\}. \quad (13)$$

It remains to prove that the unit orthogonal system $\{\Phi^{A,B}_{nm}(x,y),\ |n|, |m| = 0, 1, \ldots, \infty\}$ is complete along $[-\frac{T_1}{2}, \frac{T_1}{2}] \times [-\frac{T_2}{2}, \frac{T_2}{2}]$ when $k_1 = k_2 = 1$. In fact, Suppose that a finite function f satisfies

$$\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}}\int_{-\frac{T_2}{2}}^{\frac{T_2}{2}} f(x,y)\Phi^{A,B*}_{nm}(x,y)dxdy = 0.$$

According to the definition of the 2-D LCT, we have

$$\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}}\int_{-\frac{T_2}{2}}^{\frac{T_2}{2}} f(x,y)\Phi^{A,B*}_{nm}(x,y)dxdy = 2\pi\sqrt{\frac{b_1b_2}{T_1T_2}}\mathcal{L}_{A,B}[f]\left(n\frac{2\pi k_1b_1}{T_1}, m\frac{2\pi k_2b_2}{T_2}\right).$$

Thus

$$\mathcal{L}_{A,B}[f]\left(n\frac{2\pi k_1b_1}{T_1}, m\frac{2\pi k_2b_2}{T_2}\right) \equiv 0.$$

If $k_1 = k_2 = 1$, from Eq. (9) in Lemma 3.2, we obtain that

$$f(x,y) = \mathcal{L}_{A^{-1},B^{-1}}\left[\mathcal{L}_{A,B}[f]\left(n\frac{2\pi k_1b_1}{T_1}, m\frac{2\pi k_2b_2}{T_2}\right)\right] = 0.$$

This infers that the unit orthogonal series $\{\Phi^{A,B}_{nm}(x,y),\ |n|, |m| = 0, 1, \cdots, \infty\}$ is complete. Therefore, a finite function $f(x,y)$ can be reconstructed via Eq. (10). □

The following is a natural consequence of Theorem 3.4.

Corollary 3.4. *Let $f(x,y)$ be a finite function, $(x,y) \in \left[-\frac{T_1}{2}, \frac{T_1}{2}\right] \times \left[-\frac{T_2}{2}, \frac{T_2}{2}\right]$. Then, from the sampled values of the 2-D LCTS, the coefficients of its 2-D LCTS can be given by*

$$C^{A,B}_{nm} = 2\pi\sqrt{\frac{b_1b_2}{T_1T_2}}\mathcal{L}_{A,B}[f]\left(n\frac{2\pi b_1}{T_1}, m\frac{2\pi b_2}{T_2}\right).$$

Let f in Theorem 3.3 be an image function, and $T_1 = T_2 = 2$. Then the transform coefficient of the 2-D LCTS can be given by

$$M^{\mathrm{2DLCTS}}_{nm} = \int\int_D f(x,y)\Phi^{A,B*}_{nm}(x,y)dxdy, \quad (14)$$

with

$$\Phi^{A,B}_{nm}(x,y) = \frac{i}{2}\exp\left\{\frac{i}{2b_1}[d_1(n\pi b_1)^2 - 2x(n\pi b_1) + a_1x^2]\right\} \cdot \exp\left\{\frac{i}{2b_2}[d_2(m\pi b_2)^2 - 2y(m\pi b_2) + a_2y^2]\right\}, \quad (15)$$

where $D = \{(x,y)||x|, |y| \le 1\}$.

3.3. Image representation using the PLCT and 2-D LCTS

In this subsection, we discuss the discrete implementation of the PLCT and the 2-D LCTS on the theory of the image representation. Let $f(x,y)$ be an image with size $N \times M$ and a discrete form $g[k,l],\ k = 0, 1, \ldots, N-1,\ l = 0, 1, \ldots, M-1$. Project f onto a domain $(x_k, y_l) \in [-1, 1] \times [-1, 1]$ with

$$x_k = \frac{k - N/2 + 1}{N/2}, \; y_l = \frac{l - M/2 + 1}{M/2}.$$

Thus, the discrete forms in Cartesian coordinates of the PLCT and the 2-D LCTS can be respectively represented as

$$M_{nm}^{\mathrm{PLCT}} \doteq \frac{1}{\pi} \sum_{l=0}^{M-1} \sum_{k=0}^{N-1} K_{nm}^{A*}(x_k, y_l) f(x_k, y_l) \triangle x \triangle y, \tag{16}$$

$$M_{nm}^{\mathrm{2DLCTS}} \doteq \sum_{l=0}^{M-1} \sum_{k=0}^{N-1} \Phi_{nm}^{A,B*}(x_k, y_l) f(x_k, y_l) \triangle x \triangle y, \tag{17}$$

subject to $x_k^2 + y_l^2 \leq 1$, where K_{nm}^{A} and $\Phi_{nm}^{A,B}$ are respectively given by Eqs. (7) and (15), and $f(x_k, y_l) = g[k, l]$, $\triangle x = 2/N$, $\triangle y = 2/M$.

The computation of the transform coefficients of image function f is a decomposing process. In the following, we discuss the reconstructing process. It is known that the kernels of the PLCT and the 2-D LCTS are all orthogonal and complete. Thus, any image function f can be expressed in terms of their coefficients, as follows

$$f(r, \theta) = \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} M_{nm} K_{nm}(r, \theta). \tag{18}$$

While, in practical computation, we always consider finite n and m in Eq. (18). Let $I_K = \{(n, m) | |n| + |m| \leqslant K, K \text{is a finite number}\}$ be a subset, that is,

$$(n, m) \in I_K \subseteq \{(n, m) | |n|, |m| = 0, 1, \ldots, \infty\},$$

then an approximation of the image function can be obtained by

$$\hat{f}(r, \theta) = \sum_{(n,m) \in I_K} M_{nm} K_{nm}(r, \theta). \tag{19}$$

Consequently, from Eqs. (18) and (19), we can respectively obtain the reconstructed images of the given image by using the PLCT and the 2-D LCTS. In brief, the proposed transforms have the capabilities on the image representation in the theory.

4. Simulations

In this section, we analyze the capabilities of the PLCT and the 2-D LCTS on image representation through simulations. All of the following simulations are executed using double precision floating point values, and all the experiments are conducted on a personal computer having 2.10 GHz Processor with 1.96 GB RAM and Celeron Dual core CPU, and using MATLAB version R2013a. In the following, we always take the ZMs, the PZMs, the PCET and the PCT as standard. The reasons are as follows: (1) they have been successfully applied in the image reconstruction; (2) the PCET can be viewed as the special case of the PLCT and (3) the PCT has the best capability among the PHTs. The test images are shown in Fig. 1, where all images are gray scale images.

Fig. 1. The test images. *Note:* $|I_K|$ is equal to the total number of the transform coefficients when fixed the order K, and we call the approximation of an original image as the reconstructed image.

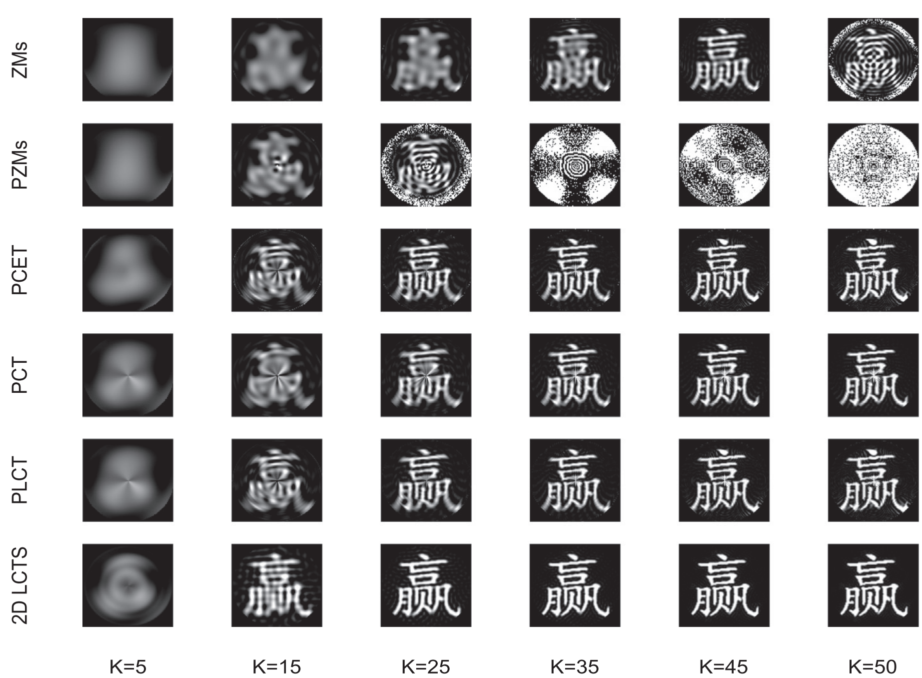

Fig. 2. Some reconstructed images of Fig. 1(a).

Table 1
Relation between the RMSE values of Fig. 1(a) and the order K.

Methods	K					
	5	15	25	35	45	50
ZM	0.676061	0.604624	0.538308	0.435116	0.391359	8.855915
PZM	0.646479	0.536357	153.8133	*	*	*
PCET	0.632306	0.49078	0.351647	0.281353	0.248905	0.235428
PCT	0.652178	0.532618	0.42858	0.33267	0.294496	0.283604
PLCT	0.624649	0.485183	0.347467	0.278727	0.245835	0.232067
2D LCT	0.620776	0.421464	0.305506	0.257291	0.206551	0.187454

* The asterisks indicate that the datas will increase with respect to K.

Table 2
Relation between the RMSE values of Fig. 1(b) and the order K.

Methods	K					
	5	15	25	35	45	50
ZM	0.750771	0.68694	0.619131	0.497739	0.317211	97.56156
PZM	0.73779	0.616581	680.9463	1.06E+18	2.47E+33	4.85E+40
PCET	0.732224	0.513594	0.306787	0.219706	0.293028	0.380193
PCT	0.737822	0.610206	0.398433	0.272422	0.232371	0.23679
PLCT	0.728261	0.505382	0.302893	0.21652	0.289069	0.373296
2D LCTS	0.764522	0.497249	0.358064	0.297367	0.236135	0.203879

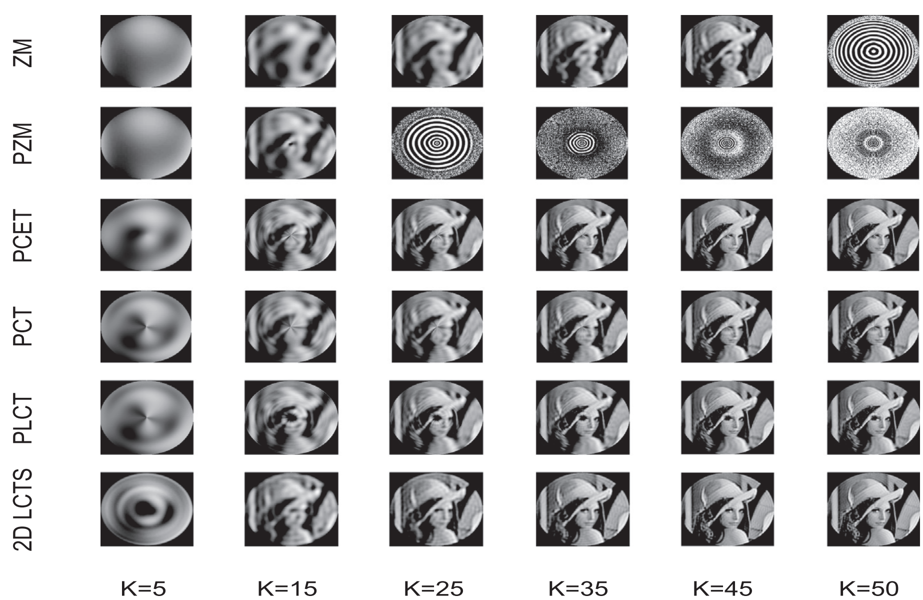

Fig. 3. Some reconstructed images of Fig. 1(c).

4.1. Image reconstruction

The capabilities of the PLCT and the 2-D LCTS on describing given images will be demonstrated via the image reconstruction. In the following, we take the contrasting characters shown in Fig. 1(b) as an example and suppose $K = 5,\ 15,\ 25,\ 35,\ 45,\ 50$, respectively, to analyze the image reconstruction. Fig. 2 shows the reconstructed images by the PLCT and the 2-D LCTS, the PCET and the PCT, respectively. It indicates that the more transform coefficients are added to the reconstruction process, the closer the reconstructed images are to the original image. We can see that the 2-D LCTS has strongest reconstruction capability among the PLCT, the PCET, the PCT, the ZMs and the PZMs. Specially, the ZMs and the PZMs have the bad reconstruction when the order K becomes bigger. Here, it needs to analyze the image reconstruction by the difference between the original image and its approximative one. Such difference is measured via Root-Mean-Square Error (RMSE):

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{x^2+y^2\le 1}\left(f(x,y)-\hat{f}(x,y)\right)^2}{\sum_{x,y} f(x,y)}}. \tag{20}$$

Taking Fig. 1(a) and (b) as example, the RMSE values determined by the six transforms are shown in Tables 1 and 2. It indicates that the 2-D LCTS has best capability on the reconstruction than the others.

We call a reconstruction is well-defined if the RMSE values of the original image and its approximation decrease along with the increase of $|I_K|$. Otherwise, the reconstruction is bad-defined.

Moreover, for sake to ensure the effectiveness and reliability of the experiments, we choose the more general test image shown in

Table 3
Relation between the RMSE values of Fig. 1(c) and the order K.

Methods	K					
	5	15	25	35	45	50
ZM	0.291138	0.207889	0.174216	0.15081	0.138822	1367.521
PZM	0.270103	0.18536	735.8739	3.71E+18	9.76E+33	9.04E+40
PCET	0.258534	0.17648	0.135815	0.115738	0.100226	0.094461
PCT	0.26636	0.189685	0.148376	0.127478	0.112076	0.105434
PLCT	0.307627	0.200277	0.152758	0.129165	0.111187	0.104716
2D LCTS	0.432582	0.309316	0.266875	0.240576	0.221591	0.213093

Table 4
The relationship between the MSE and the rotation factors.

Methods	R				
	0°	45°	90°	135°	180°
PCET	0	0.00354	0.012016	0.020275	0.018971
PCT	0	0.003743	0.008373	0.014489	0.010769
PLCT	0	0.003501	0.011247	0.01824	0.016411
2D LCTs	0	1.864572	2.424592	2.153566	0.918588
	225°	270°	315°	360°	
PCET	0.011464	0.004198	0.001691	0	
PCT	0.005585	0.001335	0.000572	0	
PLCT	0.010267	0.004224	0.002319	0	
2D LCTs	2.260259	2.438342	1.854323	0	

Table 5
The relationship between the MSE and the scaling factors.

Methods	S				
	0.1	0.2	0.3	0.4	0.5
PCET	10.42012	0.534682	0.149615	0.013711	0.00535
PCT	3.195657	0.090261	0.085092	0.008284	0.003524
PLCT	9.225122	0.414573	0.132449	0.01363	0.005749
2D LCTs	0.507254	0.079415	0.104431	0.009692	0.00446
	0.6	0.7	0.8	0.9	
PCET	0.00265	0.001234	0.000392	0.000464	
PCT	0.002268	0.00118	0.000318	0.000607	
PLCT	0.002612	0.001182	0.000383	0.00029	
2D LCTs	0.002046	0.000888	0.00034	0.000417	

Fig. 1(c) and use it for image reconstruction. Fig. 3 shows some reconstructed images and Table 3 displays the relationship between the RMSE and the $|I_K|$. The results show that the PCT is much better capability on the reconstruction with respect to the image as Lena, and the ZMs and the PZMs have the numerical stability issues.

From above results, we thus have that: (1) the novel transforms have all well-defined reconstruction; (2) in regard to the contrasting character images as in Fig. 1(a) and (b), the 2-D LCTS has strongest capability than the others on the image reconstruction. In short, we obtain that the PLCT and the 2-D LCTS have the capabilities on the image reconstruction.

4.2. Invariance analysis

In this subsection, we explore whether the PLCT or the 2-D LCTS has respectively the invariance properties of the rotation, the scale and the noise. We first consider the rotation invariance which is an important property of orthogonal moments and transforms. To explore it, take the Lena image as an example, and we can obtain the transform coefficients M^{α}_{nm} of the rotated image by the angle α. Xin et al. in [21] introduced the mean square error (MSE):

$$\mathrm{MSE} = \frac{1}{N_c} \sum_{|n|+|m| \leqslant K} \left(|M^{\alpha}_{nm}| - |M_{nm}|\right)^2, \tag{21}$$

which is a general approach to measure the invariance property. Here M_{nm} is the transform coefficient of the non-rotated image, and N_c is the total number of the transform coefficients. Table 4 is an illustration of the relationship between the MSE and the rotation factors. It also indicates that the PLCT has the property of the rotation invariance. Note that the curve corresponding to the 2-D LCTS lies above those of the PLCT. This implies that the 2-D LCTS has worse capability on the rotation invariance.

Now, we discuss the scale invariance properties of the PLCT and the 2-D LCTS, respectively. We still take the Lena image as an example. As usual, the scaling factor α ranges from 0.2 to 2 with an interval 0.2. Table 5 exhibits the relations between the MSE and the scaling factors, which implies that the 2-D LCTS provides better scale invariance than the others.

Finally, we mainly consider the noise invariance of the PLCT and the 2-D LCTS compared to those of the PCET and the PCT. The image given in Fig. 1(b) is perturbed by those Gaussian noises with variance 0.05, 0.10, 0.15 and 0.20, respectively. We have the noisy images from Fig. 4. In 1990, Khotanzad and Hong presented the average classification rate in [17]. In the following, we study the noise tolerances of the PLCT and the 2-D LCTS by using the average classification rate. The corresponding results are shown in Table 6.

In a word, we see that the PLCT has the properties of the rotation invariance and the scale invariance, while the 2-D LCTS has more strong scale invariance but worse rotation invariance. The noisy experimental results indicate that the 2-D LCTS is much better in the noise invariance than the PLCT, the PCET and the PCT.

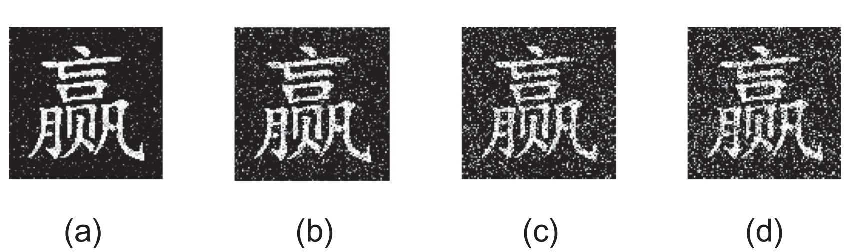

Fig. 4. The noisy images of Fig. 1(b) are added with Gaussian noise. From left to right the variance is 0.05, 0.1, 0.15, 0.2.

Table 6
Average classification rates for the given image under different degrees of gaussian noise.

σ^2	Methods	K				
		4	14	24	34	44
0.05	2-D LCT	20.820	41.387	57.514	61.382	63.915
	PLCT	19.993	36.101	51.934	56.756	53.821
	PCET	20.016	36.113	51.946	56.779	53.829
	PCT	19.564	29.404	45.908	54.114	57.214
0.1	2-D LCT	19.588	38.414	52.114	54.332	54.749
	PLCT	18.593	33.410	47.119	49.516	44.544
	PCET	18.592	33.410	47.119	49.516	44.544
	PCT	18.185	27.170	41.790	48.10	49.154
0.15	2-D LCT	18.436	35.630	47.293	47.968	47.111
	PLCT	17.39	30.858	42.375	43.080	36.394
	PCET	17.393	30.864	42.379	43.089	36.396
	PCT	16.957	25.567	39.044	44.553	44.802
0.2	2-D LCT	17.356	33.150	43.273	43.453	41.448
	PLCT	15.587	28.513	38.513	37.562	30.369
	PCET	16.007	28.573	38.612	37.744	30.433
	PCT	15.765	23.450	35.748	39.916	39.214

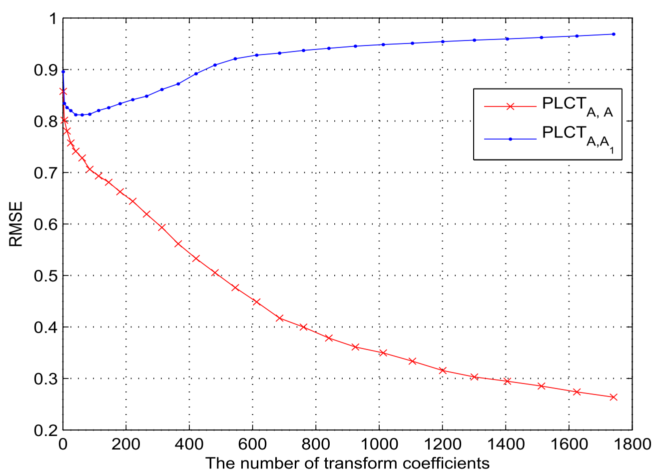

Fig. 5. Influence of consistence of parameters corresponding to the PLCT on image reconstruction.

4.3. Parameter sensitivity

Different from the transforms of PCET and PCT, it is known that the PLCT with a matrix parameter in $SL(2,R)$ has 3 free parameters including $(1,1)$-th element, $(1,2)$-th element and $(2,2)$-th element, and the 2-D LCTS having matrix parameters in $SL(2,R)$ have 6 free parameters except for $(2,1)$-th elements. It is therefore worthy and interesting to investigate the sensitivity of parameters. In the following, we analyze the parameter sensitivity by consistence of the decomposing process and the reconstructing process.

Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be respectively a decomposing process with parameter α and a reconstructing process with parameter β. Then we call $\mathcal{P}_1$ and $\mathcal{P}_2$ are consistent (also call $\mathcal{P}_1$ and $\mathcal{P}_2$ have consistence) if α equals to β.

We use $\mathcal{T}_{[\lambda_1,\lambda_2]}$ to denote a transform with the parameter λ_1 used in the decomposing process and the parameter λ_1 in the reconstructing process.

Suppose $\mathbf{A}$ and $\mathbf{A_1}$ are respectively matrix parameters in the decomposing process $\mathcal{P}_1$ and the reconstructing process $\mathcal{P}_2$ corresponding to the PLCT, seeing Table 2 for details. Fig. 5 shows that the RMSE values of $\text{PLCT}_{[\mathbf{A},\mathbf{A_1}]}$ are worse than those of $\text{PLCT}_{[\mathbf{A},\mathbf{A}]}$. It implies that the consistence of parameter in two processes can influence the transform coefficients of the PLCT (see Table 7).

Concerning on the 2-D LCTS, we need to consider the following three cases: (1) $\lambda_1 = \lambda_2 = \mathbf{AB}$; (2) $\lambda_1 = \mathbf{AB}$ and $\lambda_2 = \mathbf{CB}$ and (3) $\lambda_1 = \mathbf{AB}$ and $\lambda_2 = \mathbf{DE}$. It can be seen from Fig. 6 that the reconstruction by the $\text{PLCT2}_{[\mathbf{AB},\mathbf{AB}]}$ is well-defined, while, the 2-D $\text{LCT}_{[\mathbf{AB},\mathbf{CB}]}$ and the 2-D $\text{LCT}_{[\mathbf{AB},\mathbf{DE}]}$ have bad-defined reconstruction. Therefore, the consistency of parameters in two process is important for the image reconstruction, otherwise the transform cannot reconstruct the image. This infers that the PLCT and the 2-D LCTS with parameters have stronger capability to protect the safety of the image.

5. Discussion

Analytically, the computational cost for computing the transform coefficients based on the PLCT and the 2-D LCTS for an image of size $N * N$ is N^2 as same as the special case PCET. While, taking the PLCT as an example, we find that the PLCT has the offset form $\exp\left\{\frac{i}{2b}\left[d(2\pi nb)^2 + ar^4\right]\right\}$ which affects the radial component and the angle component, thus it is crucial to discuss the computational cost of the proposed PLCT compared with the PCET. From the simulations conducted in this paper, all the exponential factors to be used were computed and some useful factors are stored as tables in computer memory in advance for improving the speed. For example, taken the Ying image with size $300 * 300$, the times based on the PCET need 0.3839 s when the order $K = 20$; while the time based on the PLCT is about 0.3922 s at the same condition, because its radial component has more varied form due to the parameter in $SL(2,\mathbb{R})$. In addition, the form of the 2-D LCTS is more

Table 7
Parameters of the PLCT, the PLCT2 and the 2-D LCTS.

Methods	Parameters in decomposing process	Parameters in reconstructing process
$\text{PLCT}_{[\mathbf{A},\mathbf{A}]}$	$\mathbf{A} = \begin{bmatrix} 8.4072 & 2.5751 \\ 7.9135 & 2.5428 \end{bmatrix}$	(1) $\mathbf{A}$
$\text{PLCT}_{[\mathbf{A},\mathbf{A_1}]}$	$\mathbf{A}$	(2) $\mathbf{A_1} = \begin{bmatrix} 1.8351 & 6.8678 \\ 0.8390 & 3.6848 \end{bmatrix}$
2-D $\text{LCT}_{[\mathbf{AB}]}$	$\mathbf{A} = \begin{bmatrix} 8.6929 & 2.7214 \\ 18.1501 & 5.7970 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 1.4495 & 10.9972 \\ 1.0334 & 8.5303 \end{bmatrix}$	(1) $\mathbf{A}$ $\mathbf{B}$
2-D $\text{LCT}_{[\mathbf{AB},\mathbf{CB}]}$	$\mathbf{A}$ $\mathbf{B}$	(2) $\mathbf{C} = \begin{bmatrix} 3.5095 & 12.4411 \\ 1.3674 & 5.1325 \end{bmatrix}, \mathbf{B}$
2-D $\text{LCT}_{[\mathbf{AB},\mathbf{DE}]}$	$\mathbf{A}, \mathbf{B}$	(3) $\mathbf{D} = \begin{bmatrix} 1.8391 & 2.4664 \\ 1.3838 & 2.3995 \end{bmatrix}$ $\mathbf{E} = \begin{bmatrix} 0.4965 & 8.3453 \\ 0.4173 & 9.0273 \end{bmatrix}$

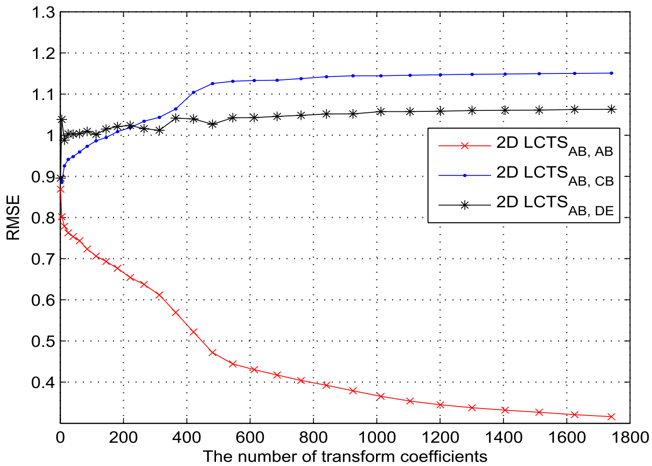

Fig. 6. Influence of consistence of parameters corresponding to the 2-D LCTS on image reconstruction.

complex than the PLCT, thus the computational time of the 2-D LCTS are more bigger than that of the PLCT.

Up to now, there is the Gaussian numerical integration method [39,40], which can improve the numerical stability and the numerical integration error, while this method needs more computational time. Thus, we will study the Gaussian numerical integration method to improve the reconstruction accuracy in the future researches.

6. Conclusion

In this paper, motivated by the PHTs and based on the theory of the LCT, we have proposed harmonic transforms with parameters in a real special linear group $SL(2, R)$ to improve the capabilities of the existing results on the image safety. The harmonic transforms include the polar harmonic linear canonical transform (PLCT) and the 2-D linear canonical transform series (2-D LCTS). The facts we derived in this paper are listed as follows: (1) the 2-D LCTS has the stronger capability than the PLCT, the PCET, the PCT, the ZMs and the PZMs on the reconstruction with respect to the contrasting characters; (2) the 2-D LCTS has much better scale invariance than other transforms; (3) compared to the PCET and the PCT without parameters, the PLCT and the 2-D LCTS have parameters in $SL(2, R)$, and if parameters used in the reconstructing process are inconsistent with those in the decomposing process, the performance of the image reconstruction can be going bad. From (3), we infer that the PLCT and the 2-D LCTS have stronger capabilities on protecting the image safety due to the many varieties of their parameters.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Nos. 61179031, 61171195 and 10932002), and is also supported by Program for New Century Excellent Talents in University (No. NCET-12-0042).

References

[1] M.K. Hu, Visual pattern recognition by moment invariants, IEEE Trans. Inform. Theory 8 (1962) 179–187.
[2] N.H. Foon, Y.H. Pang, A.T.B. Jin, D.N.C. Ling, An efficient method for human face recognition using wavelet transform and zernike moments, in: Proc. Int'l. Conf. Computer Graphics, Imaging, and Visualization, 2004, pp. 65–69.
[3] C. Kan, M.D. Srinath, Invariant character recognition with Zernike and orthogonal Fourier–Mellin moments, Pattern Recogn. 35 (2002) 143–154.
[4] M. Alghoniemy, A.H. Tewfik, Geometric invariance in image watermarking, IEEE Trans. Image Process. 13 (2004) 145–153.
[5] S. Ghosal, R. Mehrotra, Detection of composite edges, IEEE Trans. Image Process. 3 (1994) 14–25.
[6] H.S. Kim, H.K. Lee, Invariant image watermark using Zernike moments, IEEE Trans. Circ. Syst. Video Technol. 13 (2003) 766–775.
[7] J. Flusser, T. Suk, Pattern recognition by affine moment invariants, Pattern Recogn. 26 (1993) 167–174.
[8] R. Mukundan, K. Ramakrishnan, Moment Functions in Image Analysis Theory and Applications, World Scientific, 1998.
[9] H. Shu, L. Luo, J.L. Coatrieux, Moment-based approaches in imaging. Part 1: Basic features, IEEE Eng. Med. Biol. Mag. 26 (2007) 70–74.
[10] J.L. Coatrieux, Moment-based approaches in imaging. Part 2: Invariance, IEEE Eng. Med. Biol. Mag. 27 (2008) 81–83.
[11] J.L. Coatrieux, Moment-based approaches in imaging. Part 3: Computational considerations, IEEE Eng. Med. Biol. Mag. 27 (2008) 89–91.
[12] Y. Sheng, J. Duvernoy, Circular Fourier–radial Mellin descriptors for pattern recognition, J. Opt. Soc. Am. A 3 (1986) 885–888.
[13] C.H. Teh, R.T. Chin, On image analysis by the method of moments, IEEE Trans. Pattern Anal. Mach. Intell. 10 (1988) 496–513.
[14] Y.A. Mostafa, D. Psaltis, Recognitive aspects of moment invariants, IEEE Trans. Pattern Anal. Mach. Intell. 6 (1984) 698–706.
[15] Y.A. Mostafa, D. Psaltis, Image normalization by complex moments, IEEE Trans. Pattern Anal. Mach. Intell. 7 (1985) 46–55.
[16] M. Teague, Image analysis via the general theory of moments, J. Opt. Soc. Am. 70 (1980) 920–930.
[17] A. Khotanzad, Y.H. Hong, Invariant image recognition by Zernike moments, IEEE Trans. Pattern Anal. Mach. Intell. 12 (1990) 489–497.
[18] Y. Sheng, L. Shen, Orthogonal Fourier–Mellin moments for invariant pattern recognition, J. Opt. Soc. Am. A 11 (1994) 1748–1757.
[19] R. Mukundan, S.H. Ong, P.A. Lee, Image analysis by Tchebichef moments, IEEE Trans. Image Process. 10 (2001) 1357–1364.
[20] H. Ren, A. Liu, J. Zou, D. Bai, Z. Ping, Character reconstruction with radial-harmonic-Fourier moments, in: Proc. Fourth Int'l. Conf. Fuzzy Systems and Knowledge Discovery, vol. 3, 2007, pp. 307–310.
[21] Y. Xin, M. Pawlak, S. Liao, Accurate computation of Zernike moments in polar coordinates, IEEE Trans. Image Process 16 (2007) 581–587.
[22] P.T. Yap, X. Jiang, A.C. Kot, Two dimensional polar harmonic transforms for invariant image representation, IEEE Trans. Pattern Anal. Mach. Intell. 32 (2010) 1259–1270.
[23] L. Li, S. Li, A. Abraham, J.S. Pan, Geometrically invariant image watermarking using Polar Harmonic Transforms, Infrom. Sci. 199 (2012) 1–19.
[24] C. Singh, S.K. Ranade, Rotation invariant moments and transforms for geometrically invariant image watermarking, J. Electron. Imag. 22 (2013) 013034.
[25] M. Moshinsky, C. Quesne, Linear canonical transformations and their unitary representations, J. Math. Phys. 12 (1971) 1772–1783.
[26] S.A. Collins, Lens-system diffraction integral written in terms of matrix optics, J. Opt. Soc. Am. 60 (1970) 1168–1177.
[27] A. Stern, Sampling of compact signals in offset linear canonical transform domains, Signal Image Video Process. 1 (2007) 359–367.
[28] J.J. Healy, J.T. Sheridan, Sampling and discretization of the linear canonical transform, Signal Process. 89 (2009) 641–648.
[29] C.-P. Li, B.-Z. Li, T.-Z. Xu, Approximating bandlimited signals associated with the LCT domain from nonuniform samples at unknown locations, Signal Process 92 (2012) 1658–1664.
[30] Y.L. Liu, K.I. Kou, I.T. Ho, New sampling formulae for non-bandlimited signals associated with linear canonical transform and nonlinear Fourier atoms, Signal Process. 90 (2010) 933–945.
[31] J. Zhao, R. Tao, Y. Wang, Multi-channel filter banks associated with linear canonical transform, Signal Process. 93 (2013) 695–705.
[32] A. Koc, H.M. Ozaktas, C. Candan, M.A. Kutay, Digital computation of linear canonical transforms, IEEE Trans. Signal Process. 56 (2008) 2383–2394.
[33] K.B. Wolf, Integral Transforms in Science and Engineering, vol. 11, Plenum Press, NewYork, NY, USA, 1979. Chapter 9: Canonical Transforms.
[34] G.L. Xu, X.T. Wang, G.X. Xu, Uncertainty inequalities for linear canonical transform, IET Signal Process 3 (2009) 392–402.
[35] D.Y. Wei, Q.W. Ran, Y.M. Li, J. Ma, L.Y. Tan, A convolution and product theorem for the linear canonical transform, Signal Process. Lett. IEEE 16 (2009) 853–856.
[36] B.-Z. Li, R. Tao, Y. Wang, New sampling formulae related to linear canonical transform, Signal Process. 87 (2007) 983–990.
[37] R.G. Campos, J.A. Figueroa, A fast algorithm for the linear canonical transform, Signal Process. 91 (2011) 1444–1447.
[38] T.-Z. Xu, B.-Z. Li, Linear Canonical Transform and Its Applications, Science Press, Beijing, China, 2013.
[39] C. Singh, R. Upneja, Accuracy and numerical stability of high order polar harmonic transforms, IET Image Process. 6 (2012) 617–626.
[40] C. Singh, E. Walia, R. Upneja, Accurate calculation of Zernike moments, Inform. Sci. 233 (2013) 255–275.